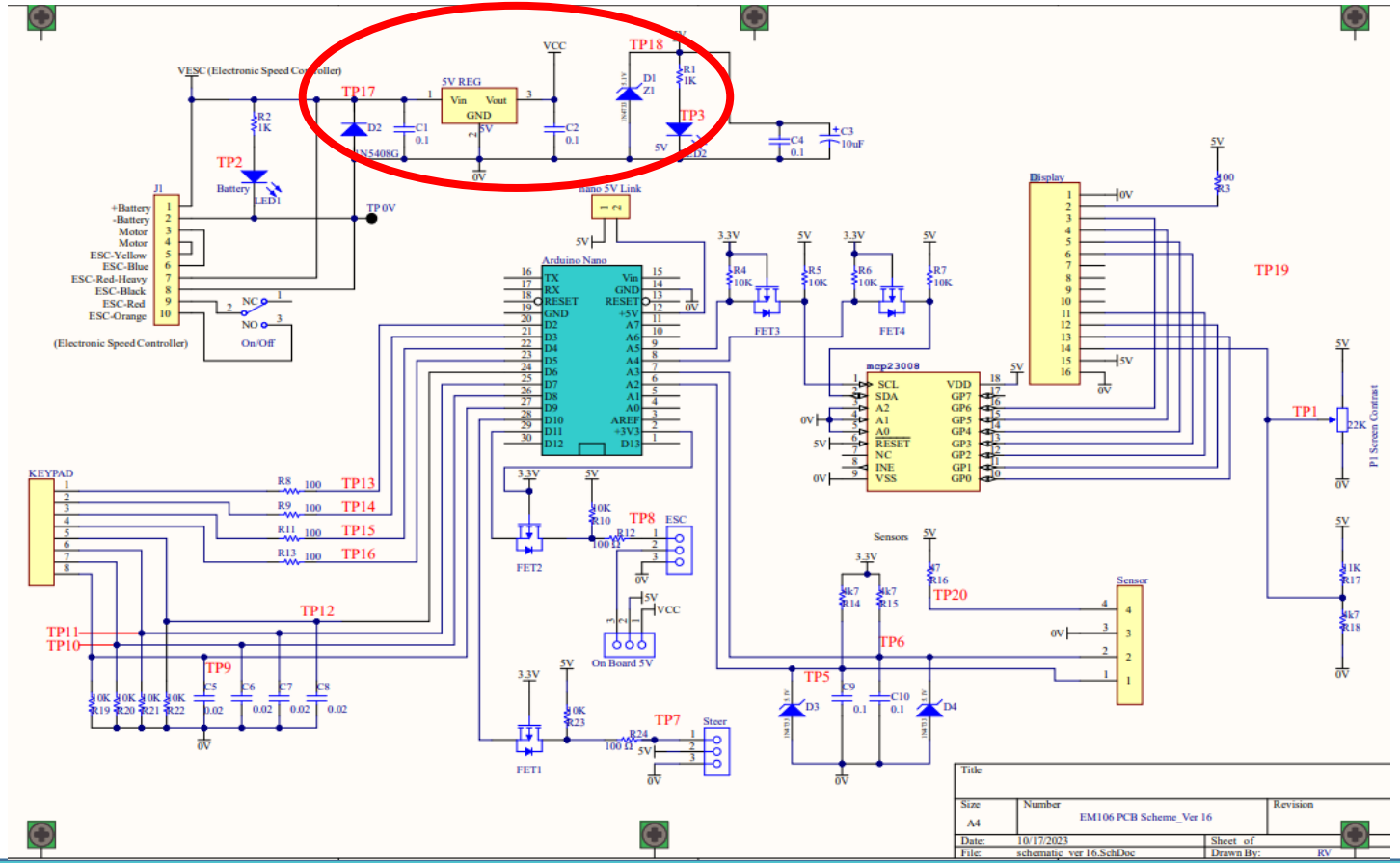
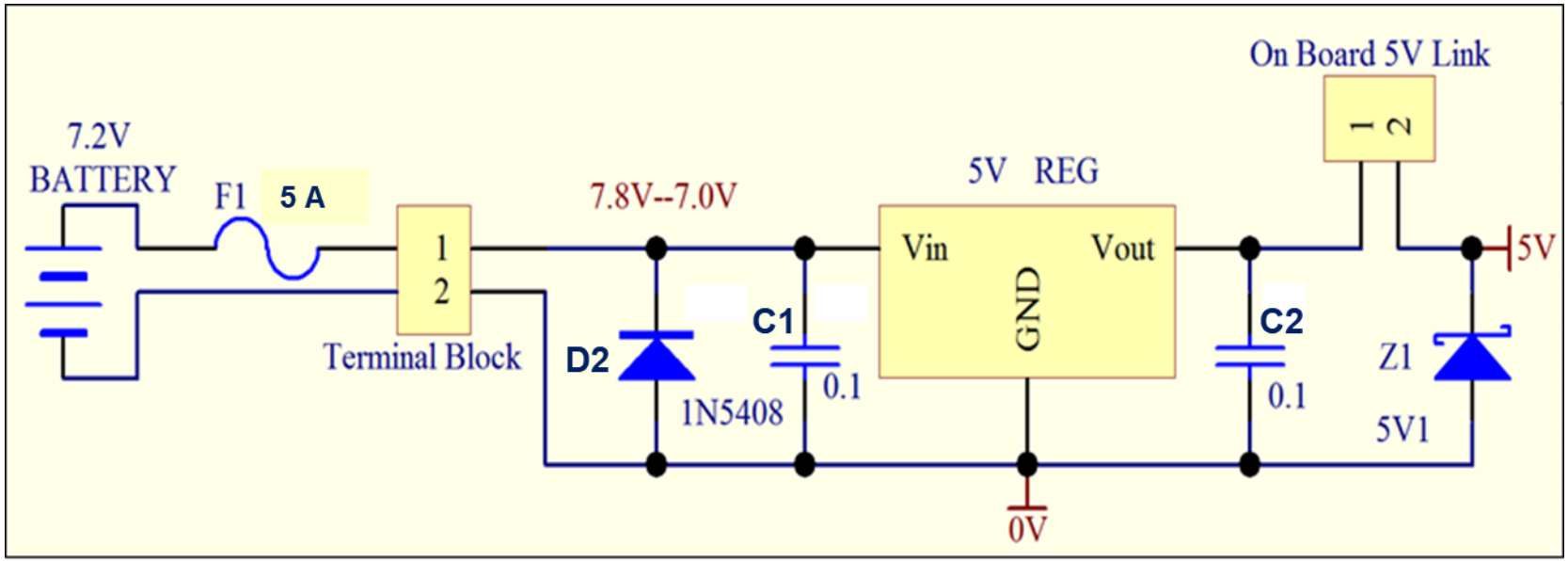
EM106 AGV Project: Arduino Nano 33_MCU

Arduino Nano 33-MCU Circuitry



Arduino Nano 33-MCU Power Supply Circuitry





5V REG

Fuse Protection

- Why?
  - Battery can deliver high currents if accidently shorted or other fault condition
  - Risk of over-heating, fire, even explosion

- Fuse rating based on max *expected* current
  - Multiply by a derating factor to avoid nuisance blowing (typically ×1.5)

- Where to locate in circuit?
  - As close as possible to the power source
  - Incorporation in either the red (positive) or black (negative) leads of the battery is equally effective
  - In provided component kit it is integrated into the red lead
  - Snap-on fuse holder for ease of replacement
  - Be careful to replace only with correct rated value

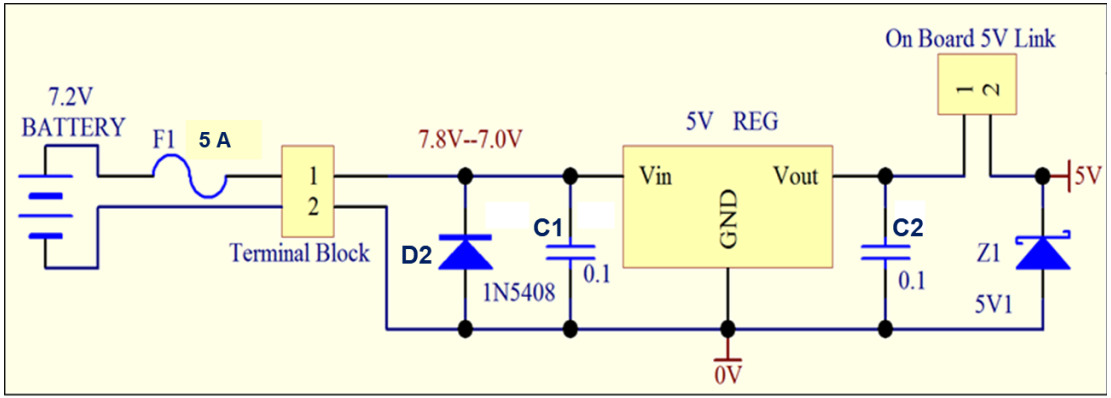


Fuse Protection

- A fuse will “blow” when excess current flows
  - Fuse wire has small, non-zero resistance, hence power dissipation: I^2R
  - As I increases, power increases, temperature increases (Heat, H= I^2Rt, t=time)
  - If temperature exceeds the melting point of the wire, the fuse fails open-circuit, disconnecting the load from the supply (battery)

- Fuses typically rated in mA or A
- For our application, *maximum* current in normal operation (say with traction motor stalled) is ~3.5A
- Nominal fuse rating: 5A

Nano 33-MCU Voltage Regulation (+Polarity Protection)

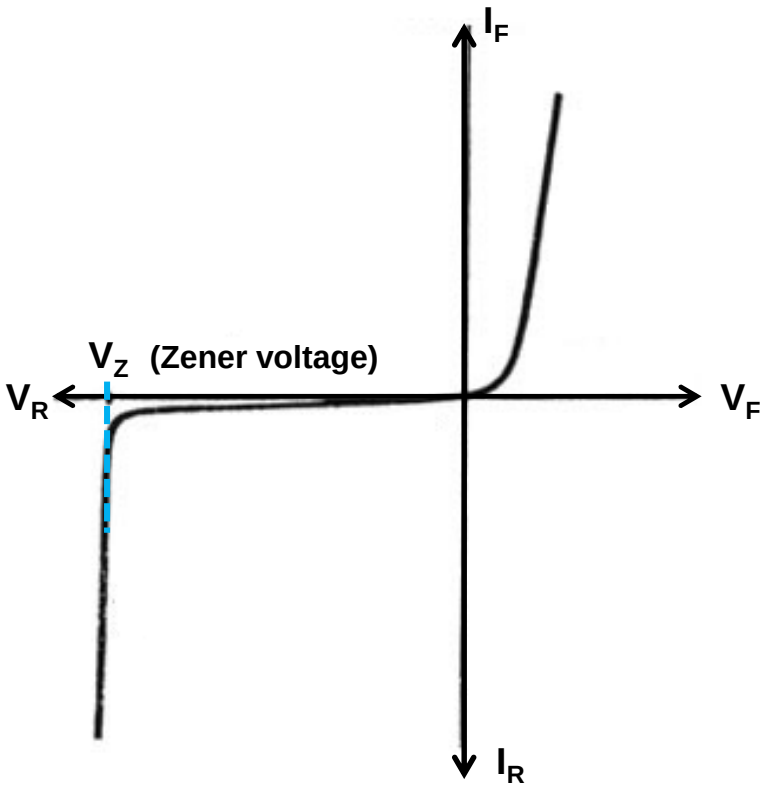


- Why do we need voltage regulation?
  - Battery voltage range is 7.8 down to 7.0 V (varies)
  - 5 V Regulator provides 5 V output voltage when battery voltage is 7.8 to 7 V.
- Diode D2 protects against *reverse connection* of battery- becoming forward biased and blowing the fuse!
- But what is the function of Zener diode Z1?

Zener Diode: Generally use as voltage reference

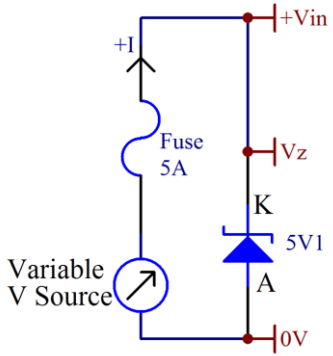
- Conventional diode: allows current only from anode to cathode
  - but current *can* also flow from cathode to anode if the reverse *breakdown voltage threshold* is reached

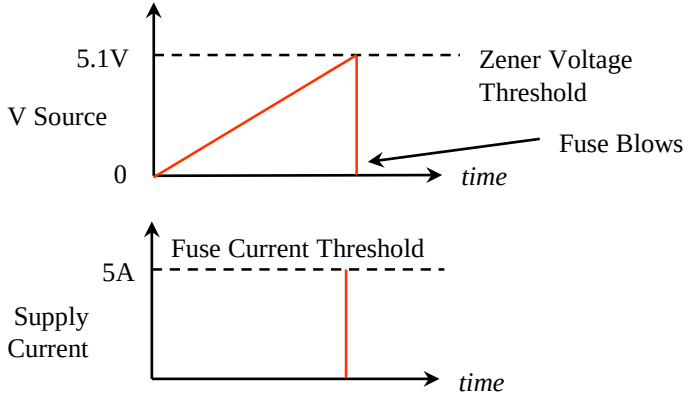
- Zener diode is *designed* to operate in reverse breakdown mode



Zener use for MCU protection?

- Here the Zener diode is to clamp the internal board voltage at a max of 5.1 V
- *Could* be exceeded if 5V regulator fails “short circuit”
- Zener diode current could rise rapidly, blowing the fuse (*because* there is no series current-limiting resistor)
- Zener diode itself may fail also - but cheaper to replace than some other components in the MCU board

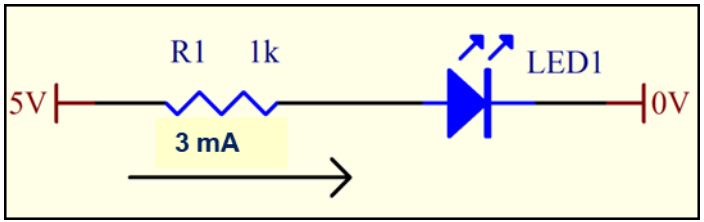




MCU Diagnostic/status LEDs

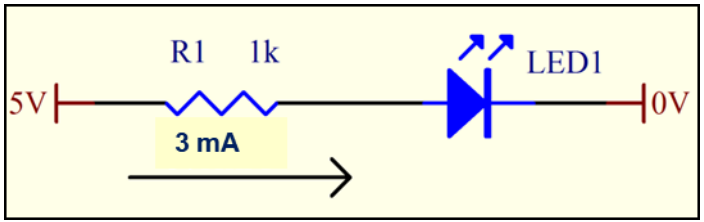
- **“Battery”**: battery is connected
- **“5 V”**: 5 V supply present
  - 5 V supply is required for the MCU, display, servo, and parts of ESC
  - Regulated down from battery voltage of ~7.0-7.8 V
  - May fail separately from failure/disconnection of battery supply
  - (When ESC is in use, 5V is provided by ESC instead of on-board regulator)

LED series resistor selection



- LEDs must be forward biased, via a *current limiting resistor*
- LED forward voltage drop is relatively insensitive to the exact current flow
  - Contrast with *resistor*
- Varies from device to device but typically ~1.9 V

LED series resistor selection



- With a 5 V supply, 3.1mA nominal current, need R1 = 1.033 kΩ?
  - Resistors manufactured in limited, discrete values
  - Most common is E12 range:
    - 1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2, with 10x multipliers
      - 0.1Ω,1Ω,10Ω,100Ω,1kΩ,10kΩ,100kΩ,1MΩ,10MΩ
      - 0.27Ω, 2.7Ω, 27Ω, 270Ω, 2.7kΩ, 27kΩ, 270kΩ, 2.7MΩ etc.
  - Our choice then would be: 1 kΩ (nominal)

Other uses of LEDs on-board the Nano33-MCU

- 2 × optical sensor module
  - Each module includes one infrared LED
  - Used to illuminate the left and right hand side of the target line
  - Reflected IR light detected by phototransistor
- LCD (Liquid Crystal Display) “backlight”: white LED